

THE ORTHOGONAL POLYNOMIALS OF THE FACTORIAL POWER SERIES
PROBABILITY DISTRIBUTIONS

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1. Abstract

In this paper methods are developed of deriving the orthogonal polynomials of the discrete factorial power series probability distribution which was defined by van Heerden (1961). Van Heerden (1961) developed methods of deriving these polynomials but in this paper these methods are adapted to the methods used by van Heerden and Gonin (1966). Recurrence formulae for the polynomials of the power series and the factorial power series distributions are also given.

2. Factorial Power Series Distribution

Van Heerden (1961) defined a discrete convergent factorial power series,

$$(1) \quad F(Z) = \sum_{x=0}^{\infty} A_x Z^{(x)}, \text{ for discrete values } x=0,1,\dots,$$

where

$$Z^{(x)} = Z(Z-1)(Z-2)\dots(Z-x+1), A_x \text{ is a constant or a real}$$

function of x and $|Z| < K$ such that,

$$A_x Z^{(x)} / F(Z) \geq 0, \text{ for all values of } x.$$

The probability that the random variable y assumes the value x , is then defined as,

$$(2) \quad P(y=x) \equiv f(x) = A_x Z^{(x)} / F(Z).$$

The factorial moments U_r of degree r of $f(x)$ is given by,

$$U_r = \sum_{x=0}^{\infty} x^{(r)} f(x)$$

$$(3) \quad = Z^{(r)} \nabla_z F(Z) / F(Z), \text{ where } \nabla_z \text{ is the receding}$$

difference operator. In terms of the factorial moments the following relation then exists,

$$(4) \quad (Z - U_1) \nabla_z U_r = U_{r+1} + U_r (r - U_1)$$

Using results (3) and (4) the following lemma can be proved:

Lemma

Let the random variable y be such that for,

$$P(y=x) = A_x Z^{(x)} / F(Z) \text{ then,}$$

$$(5) \quad a_s \nabla_z^s U_r = E \{ y^{(r)} g_s(y) \},$$

where $r, s, = 0, 1, 2, \dots$, E is the expectation operator, $g_s(y)$ a polynomial in y of degree s independent of r and a_s is defined by,

$$(6) \quad a_s = Z^{(s)} F(Z-s)/F(Z) = \sum_{i=0}^s (-1)^i \binom{s}{i} (Z-i)^{(s-i)} U_i.$$

Proof

$$\nabla_z a_s = a_s - a_{s+1}/a_1$$

and

$$\nabla_z (a_r/a_s) = \{a_r/a_s - a_{r+1}/a_{s+1}\}.$$

Assuming the truth of (5) for any r and $s = 0, 1, 2, \dots, S$ then,

$$a_1 \nabla_z (a_s \nabla_z^S U_r) = \{a_{s+1} \nabla_z^{S+1} U_r + (a_1 a_s - a_{s+1}) \nabla_z^S U_r\} \text{ and}$$

$$\begin{aligned} a_1 \nabla_z (a_s \nabla_z^S U_r) &= \{a_1 \nabla_z E(y^{(r)} g_s(y))\} \\ &= a_1 \nabla_z \{ \sum y^{(r)} g_s(y) f(y) \} \\ &= a_1 \sum y^{(r)} f(y) \{ \nabla_z g_s(y) + (g_s(y) - \nabla_z g_s(y))(y - U_1)/a_1 \} \end{aligned}$$

therefore,

$$a_{s+1} \nabla_z^{S+1} U_r = \sum y^{(r)} f(y) \{ \nabla_z g_s(y)(Z-y) + g_s(y)(y-Z + a_{s+1}/a_s) \}.$$

Therefore if there is a $g_s(y)$ for $s = 0, 1, \dots, S$, then there is a $g_{s+1}(y)$ of degree $(S+1)$ in y independent of r . This proves (5).

In particular when $r=0$ then $E\{g_s(y)\} = 0$.

The factorial power series distribution defined by (2) covers the family of hypergeometric distributions including a distribution akin to Fisher's logarithmic series distribution, used by Darwin (1960) and studied by van Heerden (1961). Van Heerden called this distribution, the Digamma Distribution.

3. Orthogonal Polynomials

If $f(x)$ is a discrete probability distribution, the orthogonal polynomial of degree r , $G_r(x)$, having the property,

$$\begin{aligned} \sum G_r(x) G_s(x) f(x) &= 0 \quad \text{for } r < s \\ &\neq 0 \quad \text{for } r = s, \end{aligned}$$

is given by Gonin (1961) in a determinantal form with functions of the factorial moments of $f(x)$ as elements. Using the method of van Heerden and Gonin (1966) the orthogonal polynomials for the factorial power series probability distribution can be derived as follows:

The orthogonal polynomial of degree r , can be written as,

$$(7) \quad G_r(x) = b_{0r} + b_{1r} x^{(1)} + b_{2r} x^{(2)} + \dots + x^{(r)}.$$

Let $g_1(x), g_2(x), \dots$ be the polynomials defined in (5) and let

$$\nabla_z^S U_r = U_r^S \text{ then,}$$

$$(8) \quad E \{g_s(x) G_r(x)\} = \sum_{j=0}^r a_s b_{jr} U_j^s, \quad s = 0, 1, \dots, r-1,$$

where $b_{rr}=1, a_0=1, U_0^0=1, U_0^i=0, U_i^0=U_i, i=1, 2, \dots, r$.

From the orthogonal property of $G_r(x)$ it follows that,

$$G_r(x) - x^{(r)} = \sum_{j=0}^{r-1} b_{jr} x^{(j)} \quad \text{and} \quad -U_r^s = \sum_{j=0}^{r-1} b_{jr} U_j^s, \quad \text{for } s=0, 1, \dots, r-1.$$

Eliminating $b_{0r}, b_{1r}, \dots, b_{r-1,r}$ it follows that,

$$(9) \quad G_r(x) = \begin{vmatrix} 1 & x^{(1)} & x^{(2)} & \dots & x^{(r)} \\ 1 & U_1 & U_2 & \dots & U_r \\ 0 & U_1^1 & U_2^1 & \dots & U_r^1 \\ 0 & U_1^2 & U_2^2 & \dots & U_r^2 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & U_1^{r-1} & U_2^{r-1} & \dots & U_r^{r-1} \end{vmatrix} \div (-1)^r |U_r^s|$$

where $|U_r^s|$ is the cofactor of $x^{(r)}$ in the numerator of (9).

From (9) it follows that,

$$G_0(x) = 1$$

$$G_1(x) = x + A_{10}/A_{11}, \quad \text{with } A_{10} = -U_1 \quad \text{and } A_{11} = 1.$$

$$G_2(x) = x^{(2)} + A_{21}x^{(1)}/A_{22} + A_{20}/A_{22} \quad \text{with,}$$

$$A_{22} = U_1^1, \quad A_{21} = -U_2^1 \quad \text{and } A_{20} = (U_1 U_2^1 - U_2 U_1^1).$$

$$G_3(x) = x^{(3)} + A_{32}x^{(2)}/A_{33} + A_{31}x^{(1)}/A_{33} + A_{30}/A_{33} \quad \text{with,}$$

$$A_{33} = (U_1^1 U_2^2 - U_2^2 U_1^1); \quad A_{32} = -(U_1^1 U_3^2 - U_3^2 U_1^1);$$

$$(10) \quad A_{31} = (U_2^1 U_3^2 - U_3^2 U_2^1); \quad A_{30} = -(U_1 A_{31} + U_2 A_{32} + U_3 A_{33}).$$

Expressions (9) and (10) are the same as that found by van Heerden and Gonin (1966) for the orthogonal polynomials of the power series probability distribution except that now,

$U_r^s = \frac{\partial}{\partial z} U_r^s$ and not equal to $\partial^s U_r / \partial z^s$, as in their case.

Put $|U_r^s|$ as defined in (9) equal to $p^{r-1}/a_1 a_2 \dots a_{r-1}$ and let

$P^r/a_1 a_2 \dots a_r$ be the cofactor of $x^{(r+1)}$ in $G_{r+1}(x)$ i.e. equal to

$|U_{r+1}^s|$ then it is well known that,

$$E \{G_s(x) G_r(x)\} = 0 \quad \text{for } s < r$$

$$= P^r/P^{r-1} = a_r A_{r+1,r+1}/A_{rr} \quad \text{for } s = r.$$

4. The First Two Orthogonal Polynomials of the Digamma Series Probability Distribution

The probability that the random variable y assumes the value x is

defined as,

$$(11) \quad P(y=x) \equiv f(x) = Z^{(x)} / xN^{(x)} F(Z), \text{ for } x=1, 2, \dots, Z,$$

where $F(Z) = \sum_{x=1}^Z Z^{(x)} / xN^{(x)} = \{D(N+1) - D(N-Z+1)\}$, with $D(Y) = d(\log_e Y)/dY$ and $|Z| < N$.

Using (3), the first two factorial moments of $f(x)$ defined in (11) are,

$$U_1 = Z/(N-Z+1)F(Z); \quad U_2 = Z^{(2)}/(N-Z+2)^{(2)}F(Z), \text{ also}$$

$$U_1^1 = \frac{Z\{(N+1)(N-Z+1)F(Z) - Z(N-Z+2)\}}{(N-Z+2)^{(2)}(N-Z+1)a_1\{F(Z)\}^2},$$

$$U_2^1 = \frac{Z^{(2)}\{2(N+1)(N-Z+1)F(Z) - Z(N-Z+3)\}}{(N-Z+3)^{(3)}(N-Z+1)a_1\{F(Z)\}^2}$$

The first two polynomials are then:

$$G_1(x) = \{x - \frac{Z}{(N-Z+1)F(Z)}\},$$

$$G_2(x) = x^{(2)} - \frac{(Z-1)\{2(N+1)(N-Z+1)F(Z) - Z(N-Z+3)\}x^{(1)}}{(N-Z+3)\{(N+1)(N-Z+1)F(Z) - Z(N-Z+2)\}}$$

$$(12) \quad + \frac{Z^{(2)}\{(N+1)(N-Z+1)\}}{(N-Z+3)^{(2)}\{(N+1)(N-Z+1)F(Z) - Z(N-Z+2)\}}.$$

5. Recurrence Formulae

It can be shown that for $f(x)$ defined by (2), $\nabla_Z f(x)/f(x)$ can be approximated by a series of polynomials which are orthogonal to $f(x)$, i.e.

$$(13) \quad \nabla_Z f(x)/f(x) = \sum_{i=0}^k C_{1i} G_i(x), \quad i=0, 1, 2, \dots, k.$$

Transposing $f(x)$ to the right-hand side of (13) and multiplying both sides by $G_k(x)$ then from (9) it follows that,

$$(14) \quad \begin{aligned} C_{1k} &= \sum_{x=0}^{\infty} \left\{ \sum_{j=0}^k A_{kj} x^{(j)} / A_{kk} \right\} \nabla_Z f(x) / \sum_{x=0}^{\infty} G_k^2(x) f(x) \\ &= \{0 + A_{k1} U_1^1 / A_{kk} + \dots + U_k^1\} / \sum_{x=0}^{\infty} G_k^2(x) f(x) \\ &= \sum_{j=1}^k A_{kj} U_j^1 / a_k A_{k+1, k+1}. \end{aligned}$$

Therefore,

$$\nabla_Z f(x)/f(x) = \{U_1^1 A_{11} G_1(x) / a_1 A_{22} + (U_1^1 A_{21} + U_2^1 A_{22}) G_2(x) / a_2 A_{33}\}$$

but,

$$\sum_{j=1}^k U_j^s A_{kj} = 0 \text{ for } s=1,2,\dots,k-1, \\ = A_{k+1,k+1} \text{ for } s=r.$$

That gives,

$$(15) \quad G_1(x) = a_1 \nabla_z f(x)/f(x)$$

In general, it can be shown that,

$$(16) \quad \nabla_z^s f(x)/f(x) = \sum_{i=0}^k C_{si} \nabla_z^i G_i(x), \text{ for } s=1,2,\dots,n \text{ and } k=n$$

By using step (14) on expression (16) for all values of s the recurrence formula for the orthogonal polynomials of the factorial power series distribution is,

$$(17) \quad G_n(x) = a_n \nabla_z^n f(x)/f(x) - \sum_{r=1}^{n-1} \frac{a_n}{a_r} G_r(x) \frac{\sum_{i=1}^r U_i^n A_{ri}}{\sum_{i=1}^r U_i^r A_{ri}}$$

Noack (1950) defined the power series probability distribution as $f(x) = A_x Z^x / F(Z)$, for $x=0,1,2,\dots$, where A_x is a constant or a real function of x , $|Z| < m$ and

$$F(Z) = \sum_{x=0}^{\infty} A_x Z^x.$$

The derivation of the recurrence formula for the orthogonal polynomials of this distribution follows the same steps as that for the polynomials of the factorial power series distribution, except that,

$\nabla_z^s f(x)/f(x)$ is replaced by $\frac{1}{f(x)} \partial^s f(x) / \partial Z^s$, $\nabla_z^s U_r$ by $\partial^s U_r / \partial Z^s$ and a_r by Z^r , then the recurrence formula is

$$(18) \quad G_n(x) = \frac{Z^n}{f(x)} \cdot \frac{\partial^n f(x)}{\partial Z^n} - \sum_{r=1}^{n-1} Z^{n-r} G_r(x) \cdot \frac{\sum_{i=1}^r U_i^n A_{ri}}{\sum_{i=1}^r U_i^r A_{ri}}$$

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